

6.104 · Software Design · MIT · Fall 2026

welcome

Mitchell Gordon & Daniel Jackson

who we are



Mitchell Gordon (Lec)



Daniel Jackson (Lec)



Eagon Meng (GTA)



Carmel Schare (GTA)



Kartik Pingle (GTA)



Arielle Blattner (Lec)



Amalia Toutziaridi (UTA)



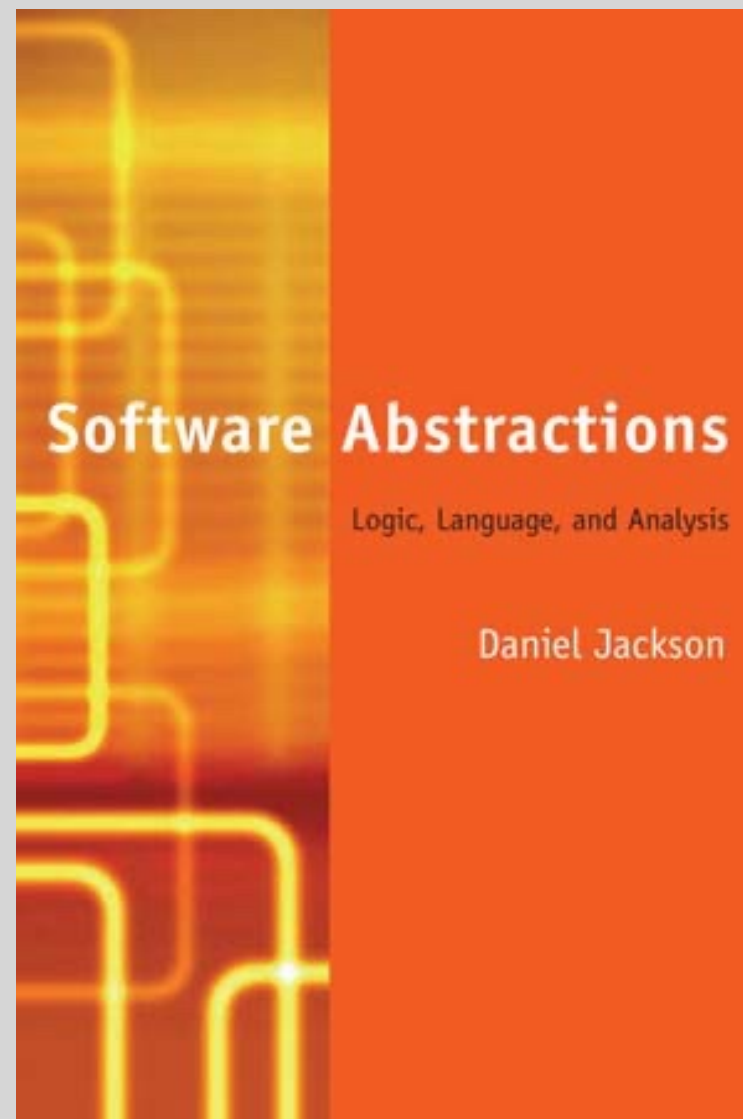
Christine Wu (UTA)



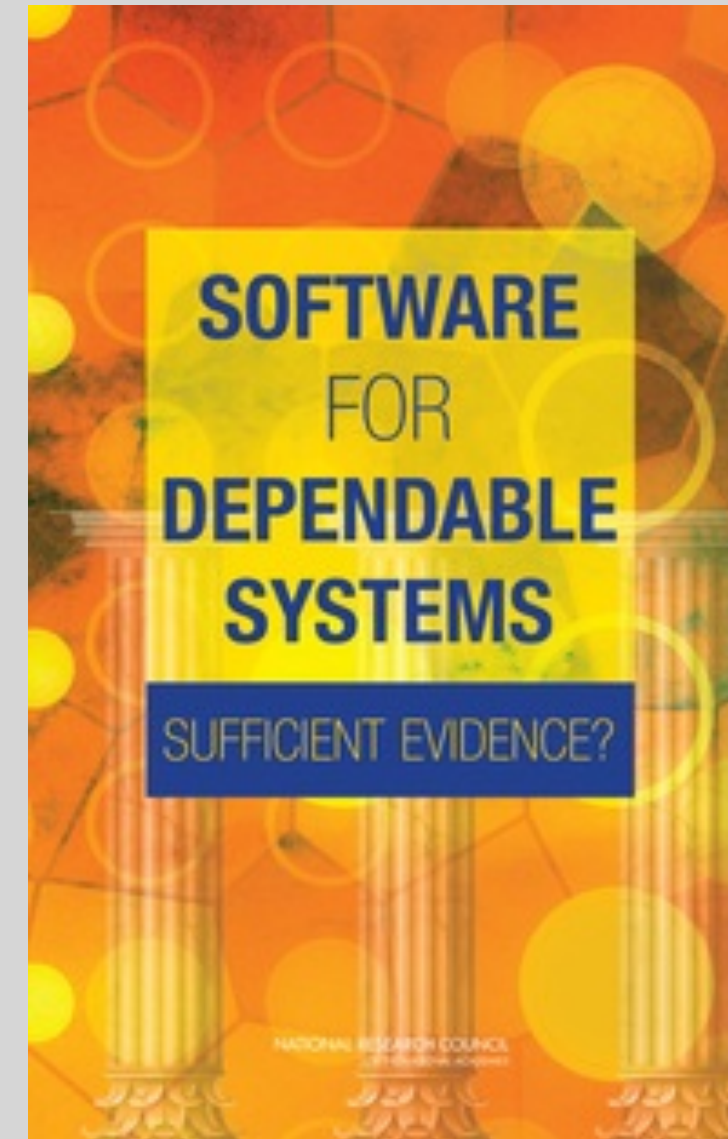
Barish Namazov (tech)

about me

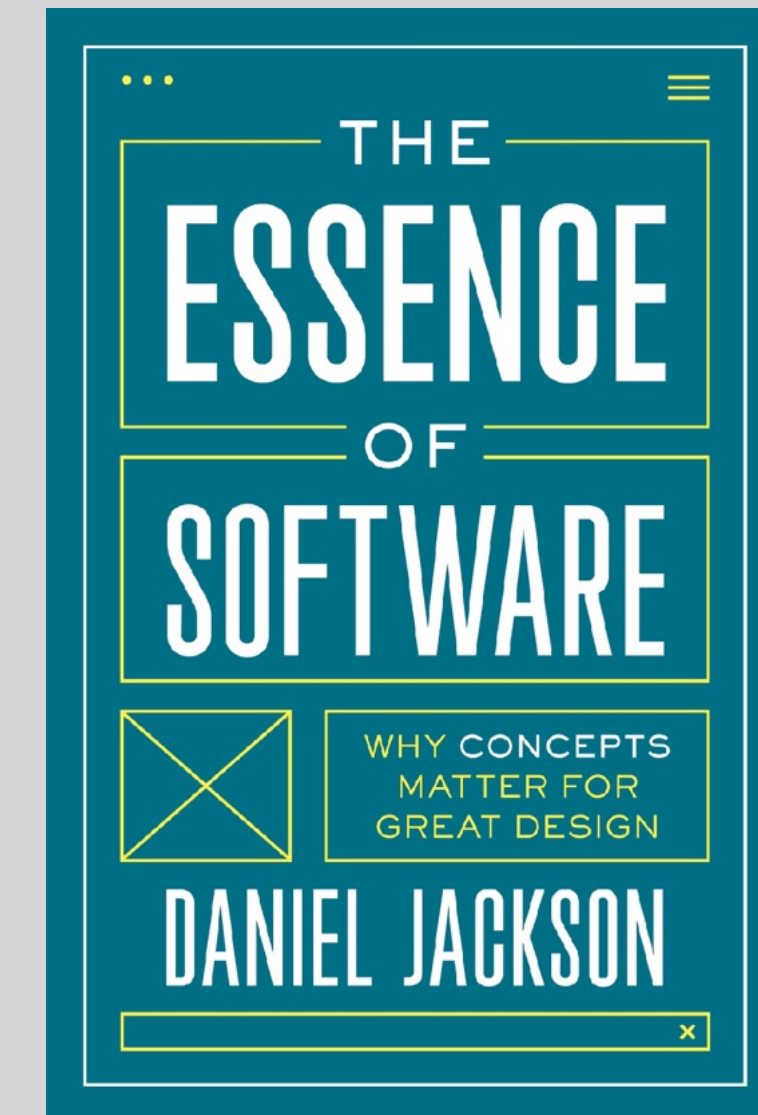
my research interests



Alloy, a modeling tool
mechanizing analysis
of software designs



dependable systems
ensuring catastrophes
don't happen



concept design
how to think about
software

when I'm not working



how about you?

what do you hope to get out of this class?



9THJXK

<https://class.mit-sdg.dev/join>

about the class

what this class is

a design class

design and build apps
using AI in a principled way

what this class isn't

a webdev class

about platforms
and all their details

a vibe coding class

code that “mostly works”
without understanding it

a UI class

about user interfaces
and visual design

an HCI class

about social &
psychological factors

this class has 2 premises

design is learnable

*not a skill people are born with
comes with practice & reflection*

AI changes everything

*design now matters more than ever
building is easier, so design can be more ambitious*

Many Computer Science (CS) programs still center around problems that AI can now solve competently. **The future belongs to developers who can model systems, anticipate edge cases, and translate ambiguity into structure—skills that AI can't automate.** We need to teach abstraction, decomposition, and specification not just as pre-coding steps, but as the new coding.



Thomas Dohmke, GitHub CEO

what's new (and isn't) in 6.104

like last year

focus on designing functionality

shaping functionality with concepts
(un)intended consequences

learning by doing on real projects

designing and building an app (2x)
working in a team (for second project)
choose your own problem

integrating LLMs

for coding and more
as app components

new this term

much improved platform

works with any coding harness
simpler structure, more reliable

more on user experience (UX)

how to conduct user testing
more on usability/ethics and AI

empowering student agency

less prescriptive assignments
skill-based rubrics
competency grading

The background of the slide is a photograph of the MIT dome, a large, circular, classical-style building with a prominent dome. The image is dark and has a reddish tint, with the text overlaid in white. The dome is the central focus, with its architectural details like columns and windows visible. The sky is a deep blue, and some trees are visible on the right side.

Report

MIT's Ad Hoc Committee on AI Use in Teaching, Learning, and Research Training

August 13, 2026

<https://aiandeducation.mit.edu/>

*this week's
activities*

complete prep (by recitation tomorrow)

setting up your GitHub repo and playing with markdown to be ready for recitation

attend recitation (tomorrow)

pick the time you prefer, can change later

join the class forum (tomorrow)

you'll get an email from us with a link

read the class guide (this weekend)

so you know how this class works and *why*

problem framing assignment (by midnight Monday)

get going early so you have time to mull it over

<http://web.mit.edu/6.104>

a word of advice

use this opportunity well

as a chance to learn

for yourself, not your grade*